Home Lab 7: Hair — Mount and Measure

Do this with a parent. Only if you have a microscope or a strong magnifier (30× or more). If you don't, do the paper homework only.

Goal

Look at real hair structure and estimate a medullary index.

You need

A microscope or strong magnifier. Clear tape, a few index cards, scissors. Hair samples: a few of your own head hairs, and — if you have a pet — a few dog or cat hairs (from a brush is fine). A clear plastic ruler with millimeter marks, or a printed micrometer scale.

Steps

1. Lay a hair straight across a window cut in an index card and tape it flat on both sides. Label the card (person / dog / cat).
2. Look at it under the scope. Find:
   - the **cuticle** — the scale pattern on the outside (flat shingles = imbricate; petals = spinous; stacked crowns = coronal)
   - the **medulla** — the darker line down the center. Is it continuous, broken up, or missing?
3. Estimate the **medulla width** and the **whole shaft width** (in the same units — even "3 ruler-ticks vs. 9 ruler-ticks" works).
4. **Medullary index = medulla width ÷ shaft width.**

Results

Hair	Cuticle pattern	Medulla (continuous / broken / absent)	Medulla width	Shaft width	MI	Human or animal?
Mine						
Pet 1						
Pet 2						

Follow-up

1. What was your hair's medullary index? Is it below 1/3 (human) or above 1/2 (animal)?
2. What cuticle scale pattern did your hair have? What about the pet hair?

3. Look at the root end of a hair you pulled vs. one from a brush. What's different? Which would be better for DNA?
4. What are the four phases of the hair growth cycle?

Coach notes

- Human: imbricate scales; thin/fragmented/absent medulla; MI well under 1/3.
- Dog: imbricate or coronal; broad continuous medulla; MI ≥ 1/2.
- Cat: spinous (petal) scales; large continuous medulla; MI > 1/2.
- Root: a pulled (anagen) hair has a soft, sometimes tissue-tagged root — good for nuclear DNA; a shed (telogen) hair has a hard club root — mitochondrial DNA only.
- Phases: anagen, catagen, telogen, exogen.